Course
Algorithms and Data Structures I

ENSM

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Chapter I

I. General Introduction to Computer Science

For all the subjects studied in college, high school or university, it is mainly a question of acquiring knowledge in different fields in order to:

1. Learn to think about various issues that are more or less complex;
2. Solve problems.

The main purpose of this approach is to understand the "scientific reasoning" and the process of "scientific knowledge production".

In mathematics, algebra and analysis are two fields that are distinguished by their content and by the types of problems they address. In high school, we have already studied arithmetic (which teaches us calculation on numbers), algebra which teaches us the resolution of equations (calculation on variables) and geometry where we learn to reason on geometric figures.

In computer science, we learn problem solving through an algorithmic approach, using algorithms. This approach is fundamentally different from mathematical proofs. If in mathematics we can be satisfied with showing the existence of an object by the proof of a theorem, in computer science, we consider that we have also to construct/find the object (the result) using an algorithm.

As the algorithms are translated into programming languages to be processed (executed) by a computer, we must also learn the architecture of the computer and how a computer works in order to allow this resolution.

But before going further, it is worth giving some definitions:

Computer Science:

- Computer science is the science of automatic information processing.
- The term *"informatique"* (in French) is a neologism proposed in 1962 by Philippe Dreyfus to characterize the automatic processing of information: It is built on the contraction of the expression *"information automatique"*. This term was accepted by the French Academy in April 1966, and "computer science" *(informatique)* then officially became the science of automatic processing of information. The word informatics does not really have an equivalent in the United States where one speaks of *"Computing Science"* or *"Computer science"* whereas *"Informatics"* is accepted by the British.

- Computing deals with two complementary aspects: The immaterial programs (software) which describe a process to be carried out and the machines (material, hardware) which execute this process.

Hardware:
Hardware is therefore all the physical elements (microprocessor, memory, screen, keyboard, hard disks, etc.) used to process data.

Software:
Software is a structured set of instructions describing information processing to be carried out by a computer hardware.

Computer :
The computer is a generic term which designates a computer equipment allowing to process information according to sequences of instructions (the programs, that are the software). The term *"ordinateur"* (in French) was proposed by the philologist Jacques Perret in April 1955 in response to a request from IBM France which considered the word "calculator" *(computer)* far too restrictive with regard to the possibilities of these machines.

The hardware architecture of a computer is based on the Von Neumann model[1]. The computer is structured around four main parts (*see* Figure 1):

1. The arithmetic/logic unit, performs the basic operations.
2. The control unit sequences the operations.
3. The memory contains both the data to be processed and the program. This will tell the control unit what calculations to make on the data.
4. Inputs-outputs are devices dedicated to communicate with the outside world (screen, keyboard, mouse, printer, scanner, etc.).

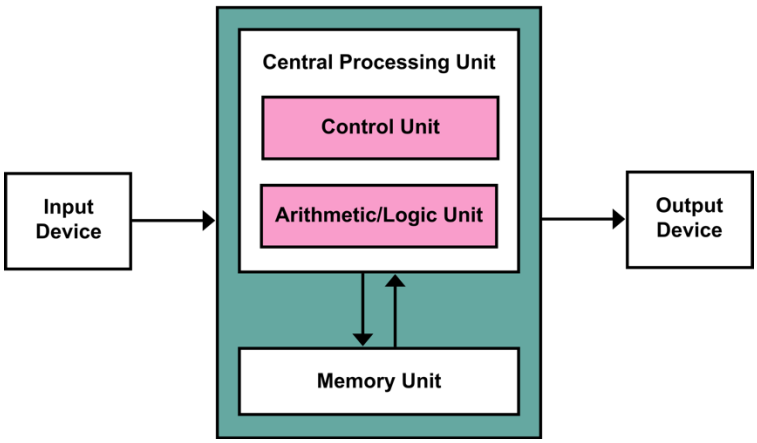


Figure 1. Von Neumann architecture.

[1] John Von Neumann (1903-1957): American mathematician of Hungarian origin. He gave his name to the von Neumann architecture used in almost all modern computers.

II. Algorithms: A brief history

a. History of the word "Algorithm"

- Algorithmics is a branch of mathematics.
- It has existed for several centuries (about 1000 years), even before the existence of the computer.
- It was born with the birth of algebra as a new mathematical discipline .
- The word Algorithm is the Latin transformation of the name of the 9th century Muslim mathematician, Muḥammad ibn Mūsa ̄al Khawārizmi ̄.

b. Why was the name of this mathematician given by his Latin successors to this discipline?

- To understand this, one must study what existed before al Khawārizmī and what his contribution brought to scientific knowledge.

c. Mathematics before al Khawārizmi ̄

Among the Greeks and their predecessors (Babylonians, Indians, etc.), there were two main mathematical disciplines:

i. Arithmetic (صناعة الحساب) : calculation on numbers (+, -, *, etc.).

ii. Euclidean geometry (صناعة الهندسة) : calculation on plane geometric figures.

In a very famous mathematical treatise that is very succinctly entitled *"al Jabr wa al Muqābala"* (الجبر والمقابلة), al Khawārizmi ̄ introduces a new mathematical discipline which allows in the same formalism to express and solve both problems of arithmetic or geometry. This new discipline consists in calculating "on variables" as one calculates on numbers. Algebra problems are expressed using expressions where the unknown to be sought is a variable which can be either of arithmetic type or of geometric type.

This new formalism makes it possible to interpret an expression of equality between two members (المقابلة/ equation). The fundamental operation between members of an equation is transposition and reduction (الجبر).

⇨ ***Refer to the solution of the quadratic equation by al Khawārizmī.***

III. The algorithm

- We can informally define the algorithm as follows:

  An algorithm is a valid computational method intended to solve a class of problems previously defined.

- An algorithm is an ordered sequence of instructions that indicates the procedure to be followed to solve a series of equivalent problems.

So when we define an algorithm, it must only contain instructions that are understandable by the person/machine who will have to execute it.

- The processing that the computer performs on the data is specified by what is called an algorithm. The algorithm describes a finite, organized, and unambiguous sequence of elementary operations in order to obtain a solution to a given problem. In general, this solution is not unique.

- In order to be executed on a computer, an algorithm has to be transformed into a program, that is to say expressed in a programming language such as C++, Python, etc.

- ***Example 1:*** A part extracted from the user manual of a fax machine explaining how to send a document.
  1. Insert the document into the automatic feeder.
  2. Dial the recipient's fax number using the dial pad.
  3. Press the send key to start the transmission.

  This user's guide explains how to send a fax. It is composed of an ordered sequence of instructions (insert…, dial…, press…) that manipulate data (document, automatic feeder, fax number, numeric keypad, send key) to perform the desired task (send a document).

- ***Example 2:*** Cooking recipe.

IV. Programming

Recall that an algorithm is an ordered sequence of instructions that indicates the approach to be followed to solve a series of equivalent problems.

Thus, it expresses the logical structure which have then to be translated into a computer language in order to produce a program. The latter is intended to be executed by a computer.

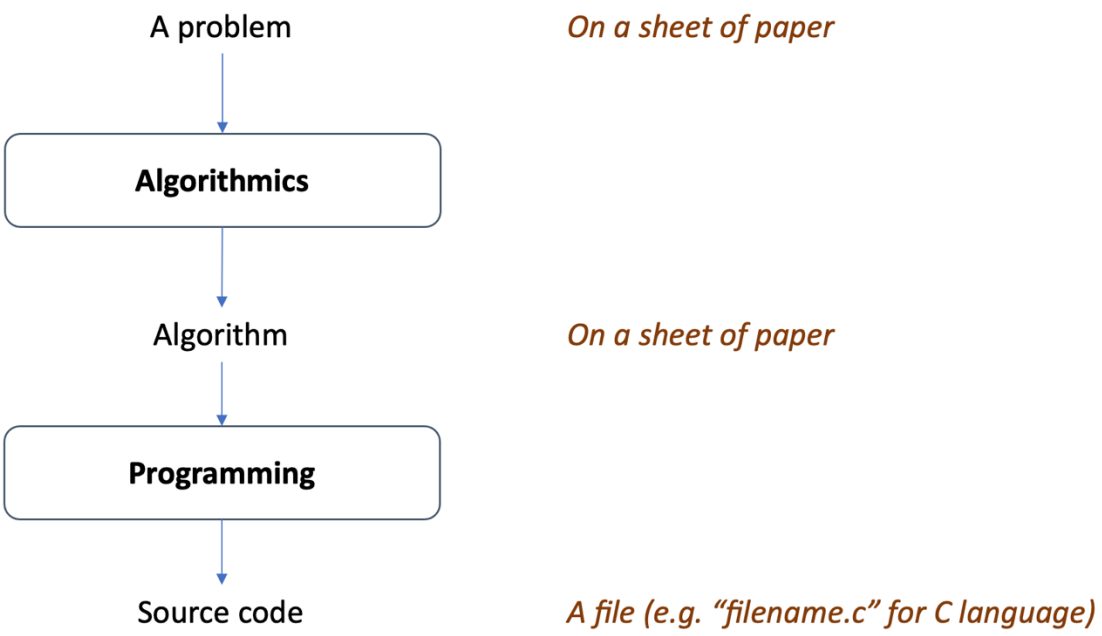
Therefore, the algorithm is independent of the programming language used to produce a computer program. On the other hand, the translation of the algorithm into a particular language depends on the chosen language.

Programming a computer consists in “explaining” in detail what it has to do, knowing that it does not “understand” human language. It makes it possible to produce a program which is nothing more than a series of instructions, encoded in very strict compliance with a set of conventions fixed in advance by a computer language.

The execution of a program on a computer then consists in decoding the instructions contained in the program by associating with each “word” of the computer language a precise action.

The program that we write in computer language using an editor is called “source program” (or “source code”).

The steps to go from a given problem to a source code are described by the diagram below:



But how does the computer make use of the source code?

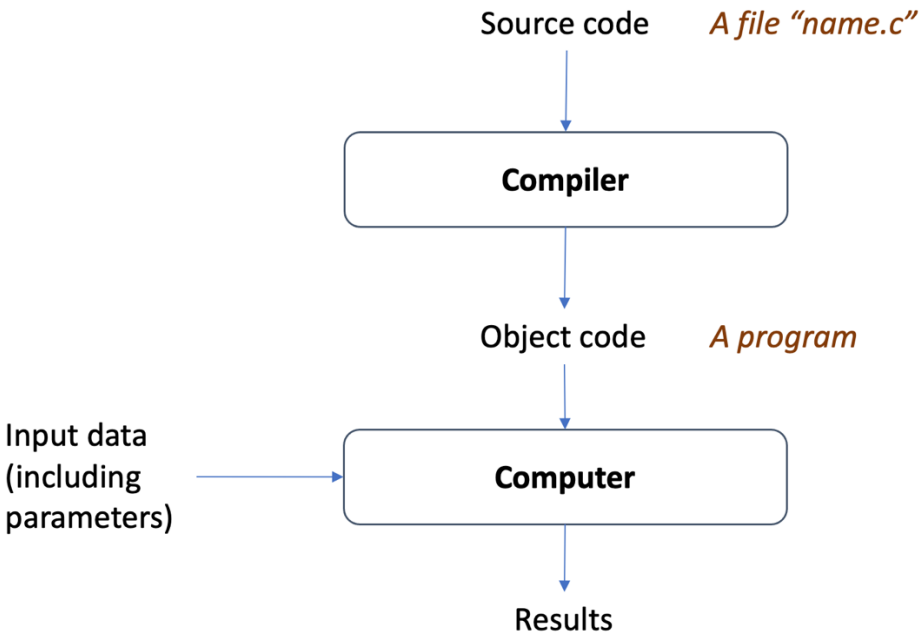
The source code undergoes a transformation into a usable/executable form by the computer, which makes it possible to obtain an "executable" program. For this, it will be necessary to use "automatic translation systems" called "compilers".

A compiler is a piece of software that translates the entire source code at once. The compiler reads all the lines of the source program, checks that they respect the programming language used and produces a new sequence of instructions called an "executable" program (or object code). This can be executed independently of the compiler and can be kept as is in a file ("executable file").

If the source code contains errors, the compiler displays error messages. It is necessary to:

- Read and understand the messages;
- Correct errors;
- Recompile the corrected source code file.

The figure below illustrates the steps for supporting a source code.


Source code
A file "name.c"
Compiler
Object code
A program
Input data
(including
parameters)
Computer
Results